

ECO102: Introduction to Macroeconomics

Lecture 11: The Money Market *(Chapter 25: Money, The Price Level, and Inflation)*

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Chapter 25: Money, The Price Level, and Inflation, pp 589 – 605

What is Money

- Any commodity or token that is generally acceptable as a means of payment is defined as money
- A ***means of payment*** is a method of settling a debt
- ***Wampum, tobacco, and nickels and dimes***
- Money serves three other functions:
 - ***Medium of exchange:*** A medium of exchange is any object that is generally accepted in exchange for goods and services
 - ***Unit of account:*** A unit of account is an agreed measure for stating the prices of goods and services
 - ***Store of value:*** Money is a store of value in the sense that it can be held and exchanged later for goods and services

Measurements of Money

- In most countries money primarily consists of:
- **Currency:** The notes and coins held by individuals and businesses are known as currency
- **Deposits:** Deposits of individuals and businesses at banks and other depository institutions, such as savings and loan associations, are also counted as money

Measures of Money

- Two official measures of money are known as M1 and M2.

M1 consists of currency and traveler's checks plus checking deposits owned by individuals and businesses. M1 does not include currency held by banks, and it does not include currency and checking deposits owned by the government

M2 consists of M1 plus time deposits, savings deposits, and money market mutual funds and other deposits

Measurements of Money

- Money is the means of payment, so the test of whether an asset is money is whether it serves as a means of payment
- Both currency and checking deposits are money because they can be transferred from one person to another by writing a check or using a debit card; meaning M1 is money
- Some of the savings deposits in M2 are just as much a means of payment as the checking deposits in M1
- Some savings deposits are not means of payment, these deposits are known as liquid assets. **Liquidity** is the property of being easily convertible into a means of payment without loss in value
- Since the deposits in M2 can quickly be converted into a means of payment they are counted as money
 - ***ARE DEPOSITS, CHECKS and CREDIT CARDS MONEY?***

The Federal Reserve's Policy Tools

- The Fed influences the quantity of money and interest rates by adjusting the quantity of reserves available to the banks and the reserves the banks must hold
- The Fed utilizes three tools to achieve its goals, they are:
- ***Open Market Operations:*** An open market operation is the purchase or sale of securities by the Fed in the loanable funds market

When the Fed buys securities, it pays for them with newly created bank reserves. When the Fed sells securities, the Fed is paid with reserves held by banks.

By changing the quantity of bank reserves, the Fed changes the quantity of monetary base, which influences the quantity of money

The Federal Reserve's Policy Tools

- ***Last Resort Loans:*** The Fed is the ***lender of last resort***, which means that if a bank is short of reserves, it can borrow from the Fed. But the Fed sets the interest rate on last resort loans and this interest rate is called the ***discount rate***
- ***Required Reserve Ratio:*** The required reserve ratio is the minimum percentage of deposits that depository institutions are required to hold as reserves.

If the Fed requires the banks to hold more reserves, the banks must cut their lending

The Money Market

- There is no limit to the amount of money we would like to receive in payment for our labour or as interest on our savings
- But there is a limit to how big an inventory of money we would like to hold and neither spend nor use to buy assets that generate an income
- The quantity of money demanded is the inventory of money that people plan to hold on any given day. It is the quantity of money in our wallets and in our deposit accounts at banks
- The quantity of money held must equal the quantity supplied, and the forces that bring about this equality in the money market have powerful effects on the economy

Influences on Money Holding

- The quantity of money that people plan to hold depends on four main factors:
- ***The price level:*** The quantity of money measured in dollars is ***nominal money***. The quantity of nominal money demanded is proportional to the price level

If the price level rises by 10 percent, people hold 10 percent more nominal money than before, other things remaining the same

The quantity of money measured in constant dollars (for example, in 2005 dollars) is ***real money***. Real money is equal to nominal money divided by the price level and is the quantity of money measured in terms of what it will buy. The quantity of real money demanded is independent of the price level

Influences on Money Holding

- ***The nominal interest rate:*** The higher the opportunity cost of holding money, other things remaining the same, the smaller is the quantity of real money demanded.

The nominal interest rate on other assets minus the nominal interest rate on money is the opportunity cost of holding money

The interest rate that you earn on currency and checking deposits is zero. So the opportunity cost of holding these items is the nominal interest rate on other assets such as a savings bond or Treasury bill

Influences on Money Holding

- **Real GDP:** The quantity of money that households and firms plan to hold depends on the amount they are spending. The quantity of money demanded in the economy as a whole depends on aggregate expenditure, i.e. real GDP
- **Financial innovation:** Technological change and the arrival of new financial products influence the quantity of money held. Financial innovations include:
 1. Daily interest checking deposits
 2. Automatic transfers between checking and saving deposits
 3. Automatic teller machines (ATMs)
 4. Credit cards and debit cards
 5. Internet banking and bill paying

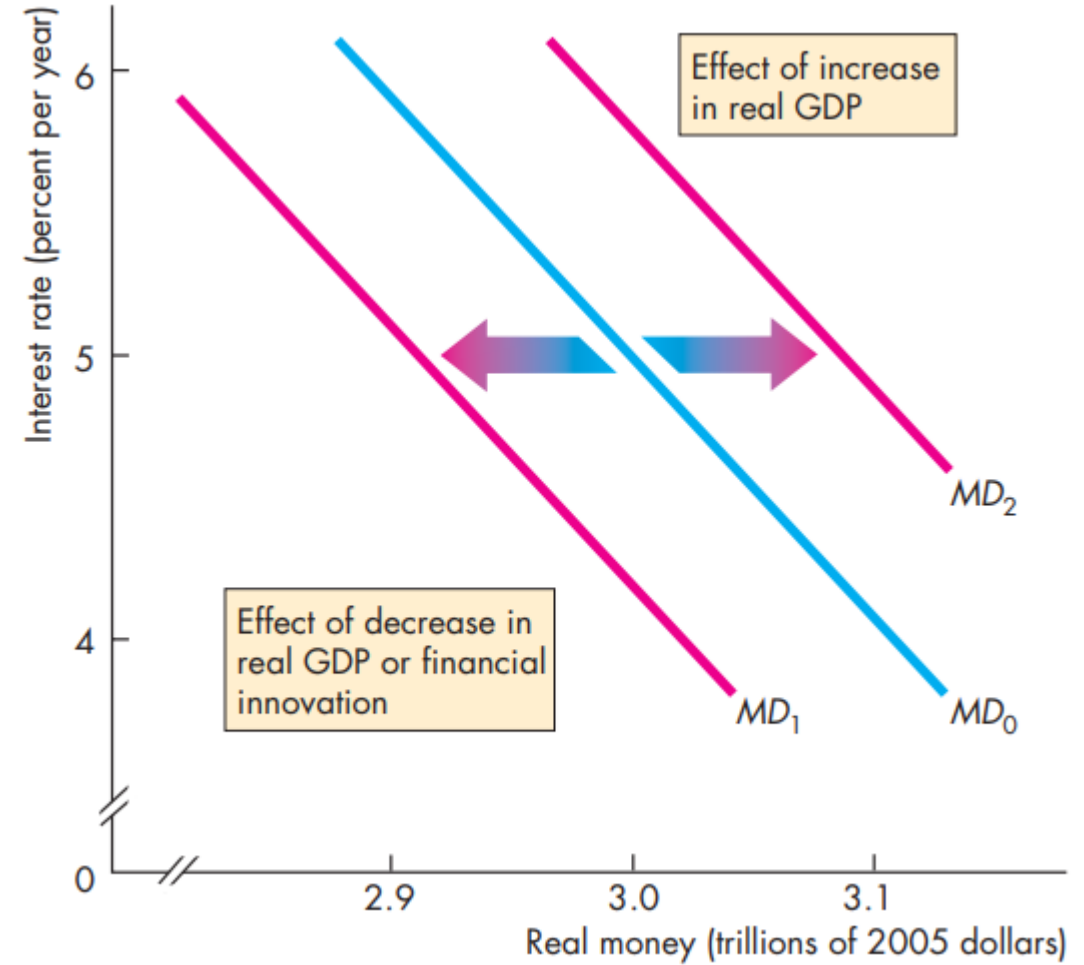
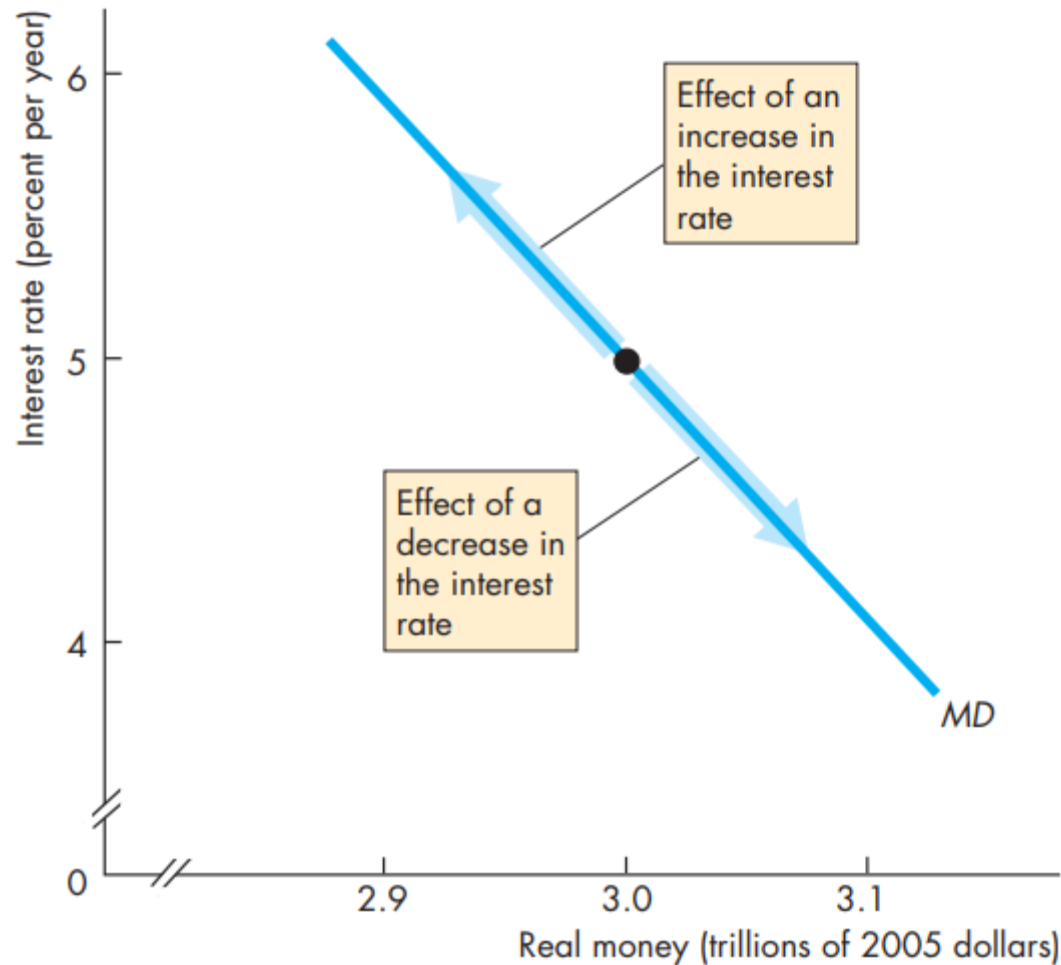
The Demand for Money

- The demand for money is the relationship between the quantity of real money demanded and the nominal interest rate when all other influences on the amount of money that people wish to hold remain the same

Shifts in the Demand for Money Curve

- A change in real GDP or financial innovation changes the demand for money and shifts the demand for money curve
- A decrease in real GDP decreases the demand for money and shifts the demand for money curve leftward; increase in real GDP has the opposite effect
- The influence of financial innovation on the demand for money curve is more complicated. But generally, financial innovation decreases the demand for money

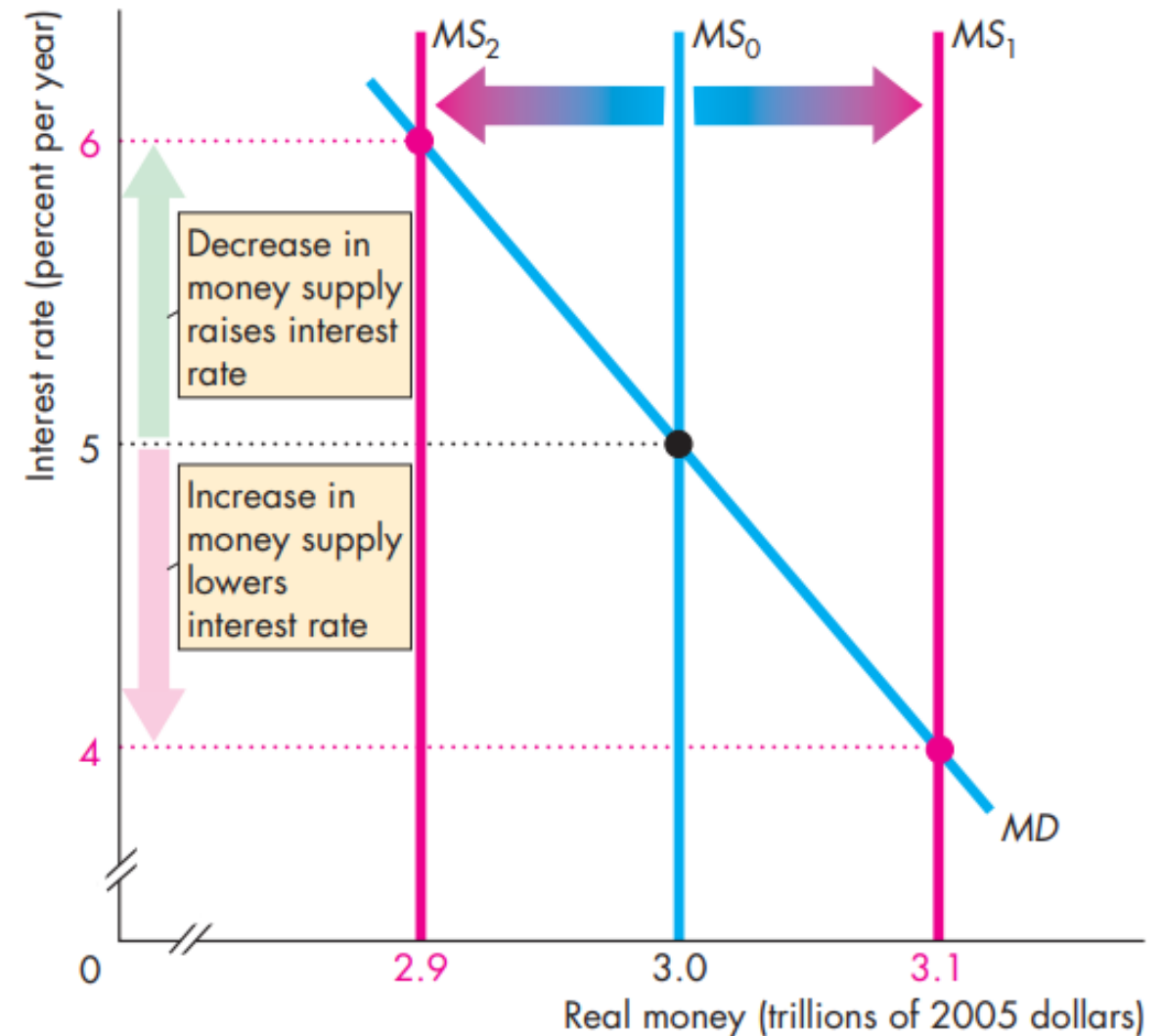
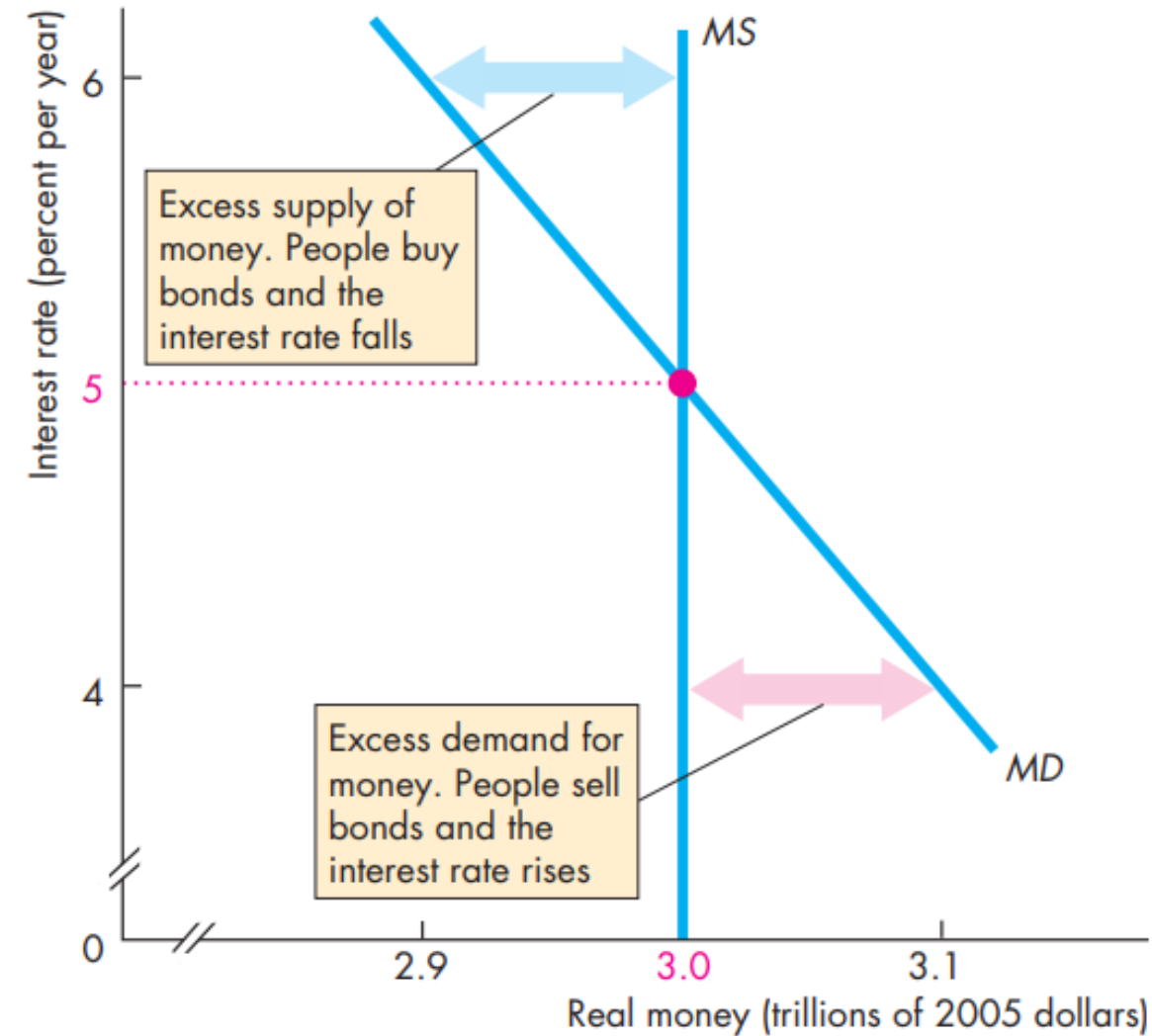
The Demand for Money



Money Market Equilibrium

- Money market equilibrium occurs when the quantity of money demanded equals the quantity of money supplied. The adjustments that occur to bring money market equilibrium are fundamentally different in the short run and the long run
- ***Short-run Equilibrium:*** Starting from a short-run equilibrium, if the Fed increases the quantity of money, people find themselves holding more money than the quantity demanded. With a surplus of money holding, people enter the loanable funds market and buy bonds. The increase in demand for bonds raises the price of a bond and lowers the interest rate.
- The opposite happens, when the Fed decreases the quantity of money

Money Market Equilibrium



Money Market Equilibrium

- **Long-run Equilibrium:** When the inflation rate equals the expected (or forecasted) inflation rate and when real GDP equals potential GDP, the money market, the loanable funds market, the goods market, and the labor market are in long-run equilibrium, meaning the economy is in long-run equilibrium
- If in long-run equilibrium, the Fed increases the quantity of money, eventually a new long-run equilibrium is reached in which none of the “real” values have changed. Real GDP, employment, the real quantity of money, and the real interest rate all return to their original levels. This happens because the price level rises by the same percentage as the rise in the quantity of money